SECTION 06 · TIME SERIES CLUSTERING

Time Series Clustering on ECG Data

Group heartbeats by shape: why Euclidean distance fails on shifted signals, and how k-Shape and HDBSCAN fix it.

aeon | k-Shape | SBD | DTW | HDBSCAN | UCR archive

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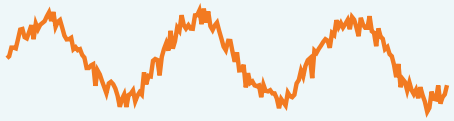
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Time Series Everywhere

Heartbeats



Machine sensors



Daily energy load



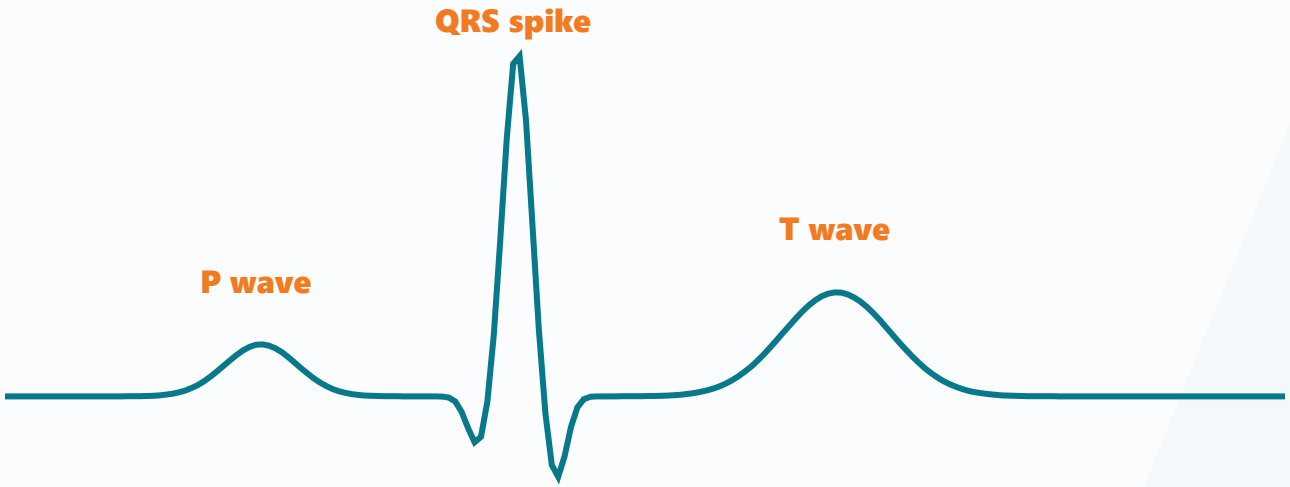
App activity



Each item is a whole sequence of readings, not one row of features.

Clustering groups sequences by shape: similar beats, similar machines, similar days.

Reading a Heartbeat Trace



one beat: about 140 readings in the UCR files

An electrocardiogram (ECG)
records the heart's electrical activity

The recording is cut into single beats, each lined up around its spike

Abnormal beats change shape
a wider spike, a missing wave,
a T wave that flips

Time Series as a 3D Array



aeon stores a collection as

`(beats, channels, readings)`

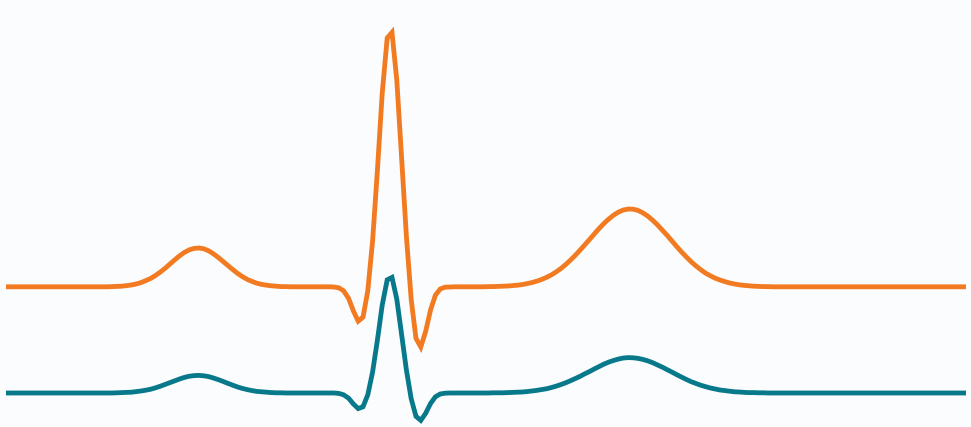
One channel here: a single ECG lead.
A wearable with three sensors would have three.

Two views of the same data

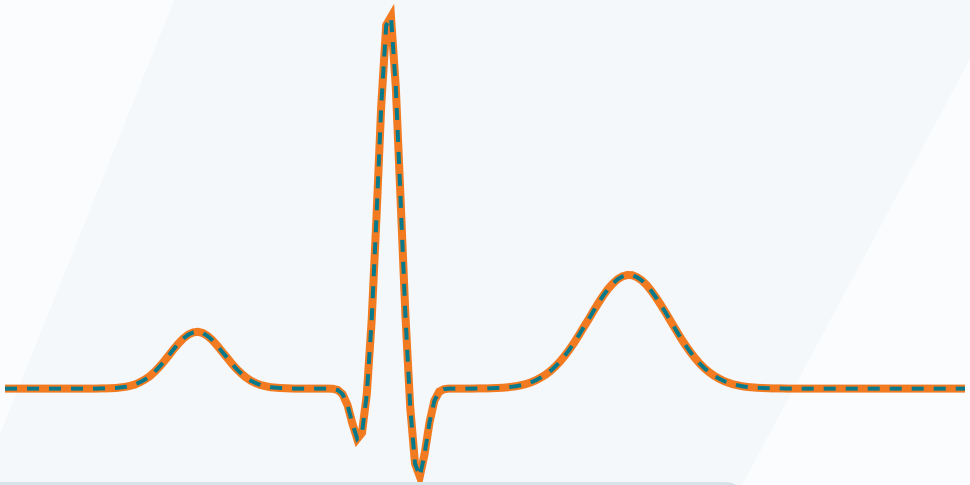
X: the 3D array aeon's clusterers expect
X[:, 0]: a plain 2D table for scikit-learn

Z-Normalising Each Series

Before: different height and offset



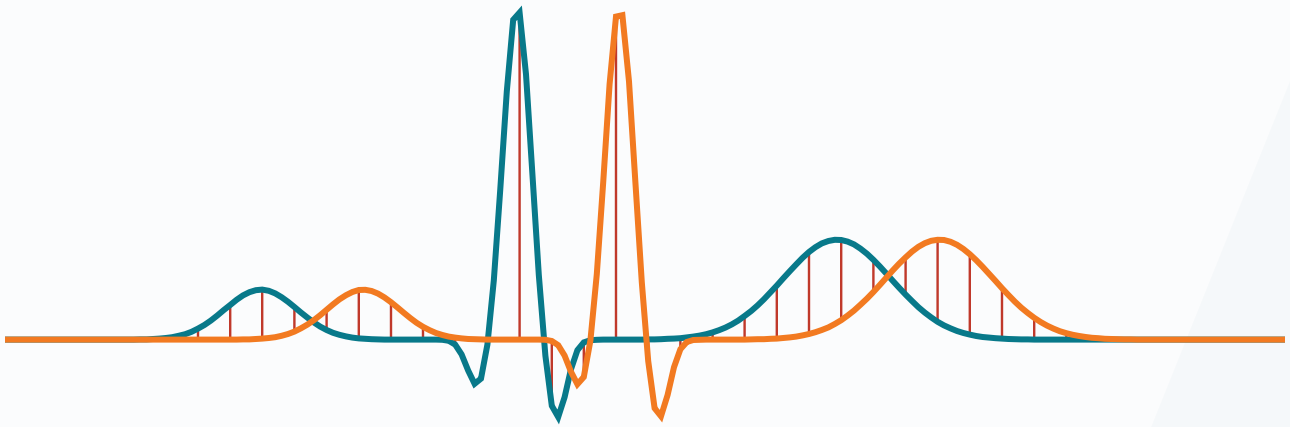
After: same shape, same scale



Subtract each series' own mean, divide by its own standard deviation

comparisons then see shape only; UCR files arrive already normalised

Why Euclidean Distance Fails on Shifts



red lines: the reading-by-reading gaps Euclidean distance adds up

Same beat, a little later

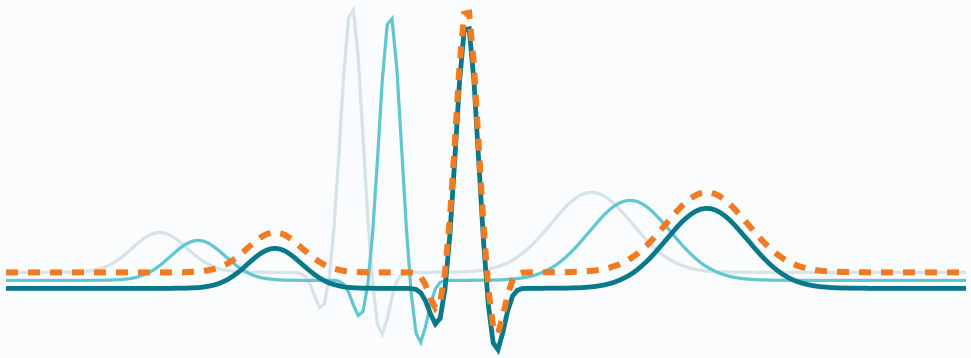
Euclidean distance compares reading 1 with reading 1, reading 2 with reading 2

The spikes miss each other, so the gap is huge where the shape is identical

A shifted copy can look further
away than a different beat

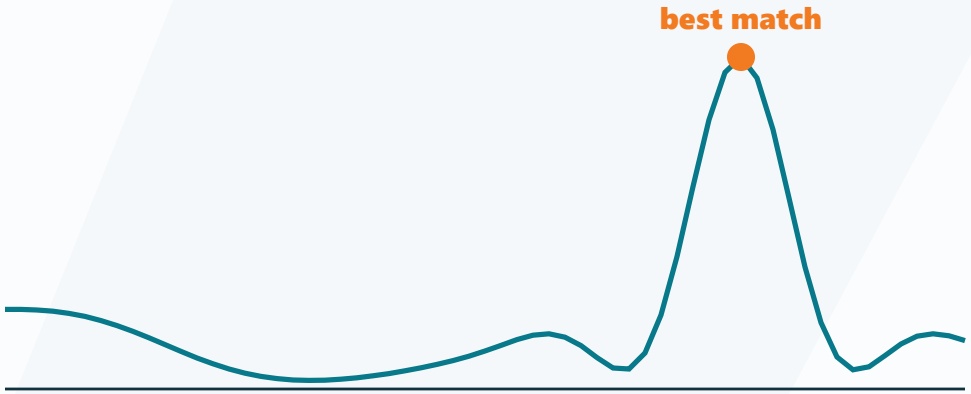
Shape-Based Distance

Slide one series over the other



at the right slide the two shapes line up

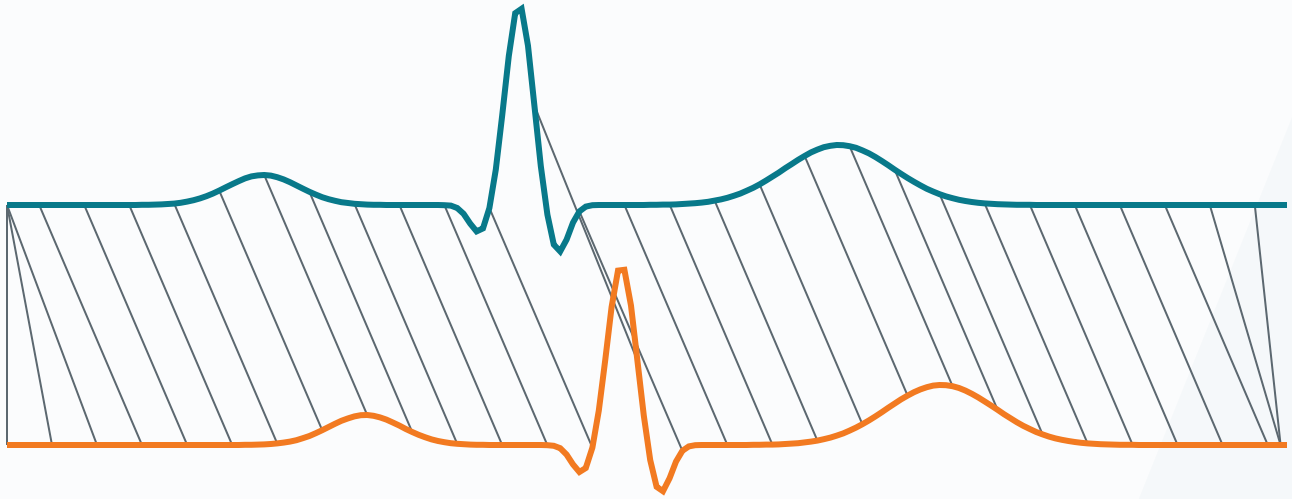
Cross-correlation at every slide



Shape-based distance (SBD) = 1 - the best normalised cross-correlation

0 for the same shape at any shift, up to 2 for opposite shapes

Dynamic Time Warping



each line pairs a reading with its best partner, not its same-index reading

Dynamic time warping (DTW)

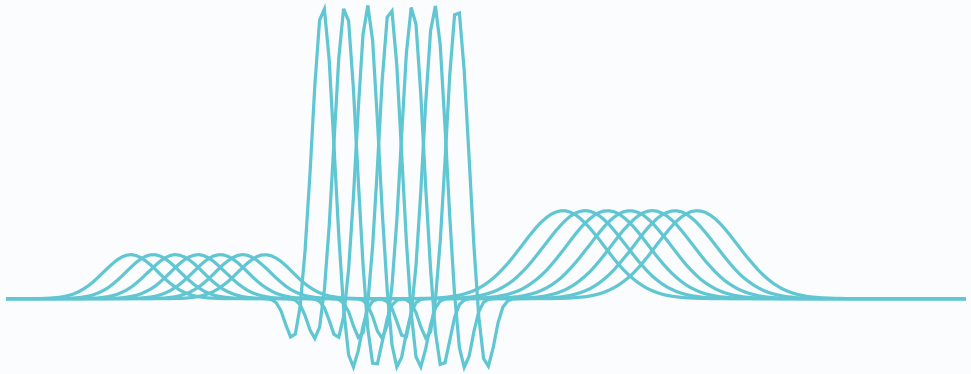
Lets one series stretch and squeeze in time to line up with the other

Handles shifts and uneven speed, like a slow beat next to a fast one

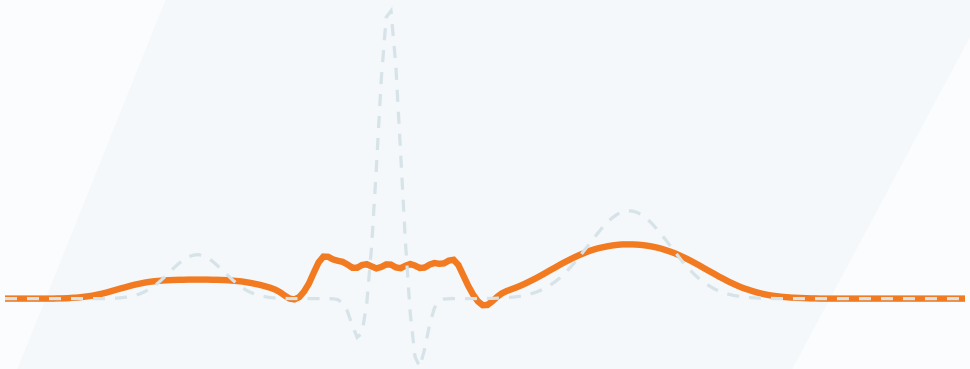
The catch: it is slow
a full table of pairings for every pair of series compared

Why k-Means Fails on Shifted Beats

Beats in one cluster, at different shifts



Their average: a blurred centre



k-means averages reading by reading, so shifted spikes smear flat

the centre looks like no real beat, and distances to it mean little

How k-Shape Works

1 Start

random groups of series

→

2 Assign

each series to the centre with the smallest SBD

→

3 Extract shape

align the group's series to the centre, then take the dominant shape

→

4 Repeat

until the groups stop changing

k-means with two parts swapped

SBD in place of Euclidean distance, so shifted beats join the same group

an aligned centre in place of a plain average, so the centre stays sharp

HDBSCAN on Raw Readings

Shifted beats spread out thinly



density is low everywhere but in tight groups

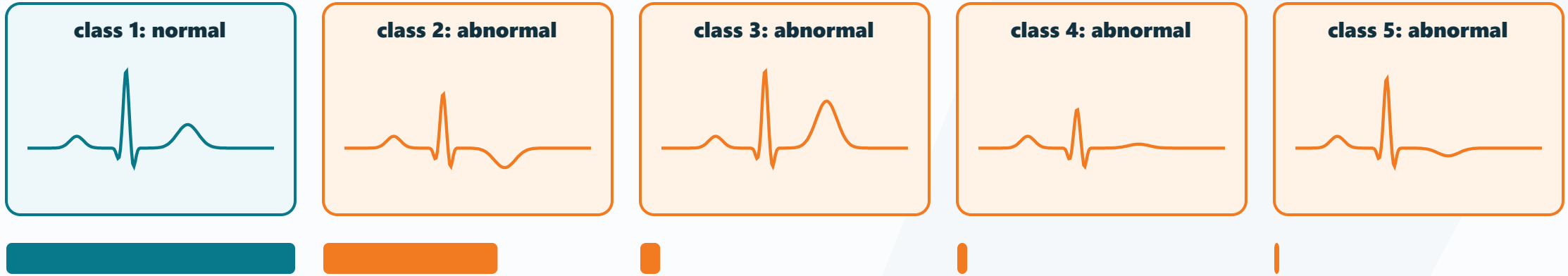
HDBSCAN looks for dense regions in the space of raw readings

When beats drift in time, copies of one shape scatter, so it finds many small groups and noise

Works well when beats are aligned

the next dataset lines every beat up on its spike, and density does much better

The ECG5000 Dataset

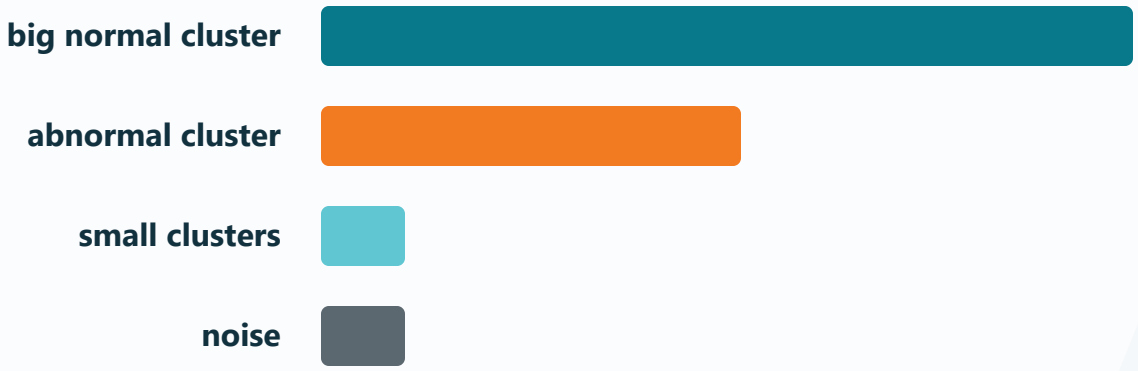


bar length: how many beats of each class, roughly

One patient, many beats, all lined up on the spike

most beats are normal; the rare classes are the ones a doctor cares about

Grading Clusters on Imbalanced Classes



what a density method tends to report

Adjusted Rand index (ARI)

agreement between clusters and the held-back classes, corrected for chance

Dominated by the big classes: getting normal beats right counts the most

Read the crosstab too

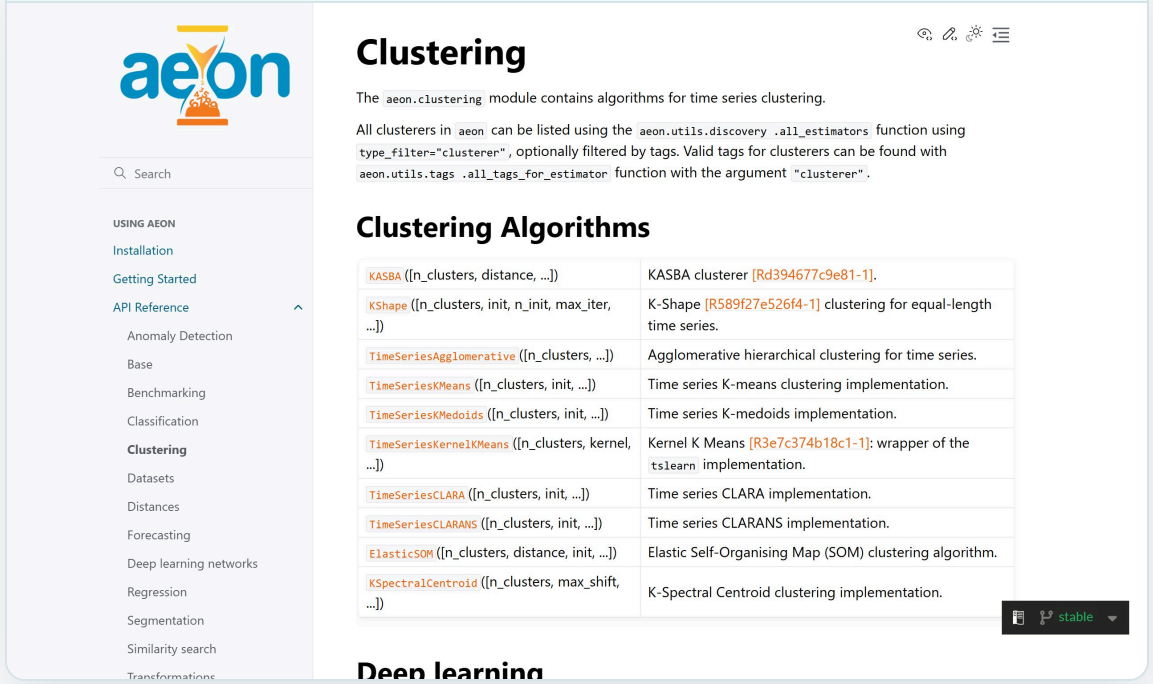
it shows where the rare classes went

Choosing a Distance

DISTANCE	IGNORES	COST	USE IT WHEN
Euclidean	nothing: reading by reading	fastest	series are already aligned
SBD	shifts in time, scale	fast, via the FFT	the same shape can start at different times
DTW	shifts and uneven speed	slow, grows with length squared	short series that stretch, like gestures or speech

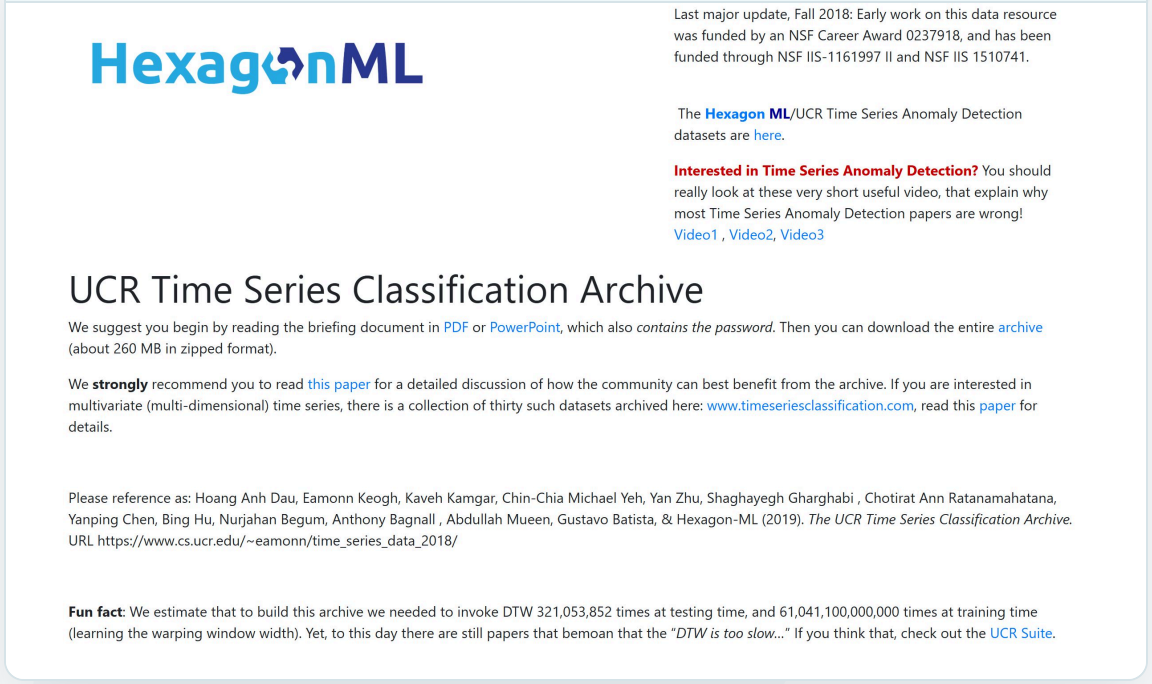
aeon Docs and the UCR Archive

aeon-toolkit.org



aeon: k-Shape, DTW k-means and other time series clusterers

cs.ucr.edu/~eamonn/time_series_data_2018



The UCR archive: 128 labelled time series datasets

aeon downloads UCR datasets by name and caches them, so no manual download is needed.

Where Time Series Clustering Is Used

Patient monitoring

group beats, flag rare shapes for a cardiologist to review

Predictive maintenance

group vibration patterns, spot a machine drifting away

Energy

group daily load curves into customer profiles

Product analytics

group weekly usage curves into engagement types

Usually a first step

clusters become labels for a person to name, or a baseline to watch for anomalies

Choosing a Clustering Method

METHOD	DISTANCE	NEEDS K	STRENGTH	WEAK SPOT
k-means	Euclidean	yes	fast, simple	blurs shifted series
k-Shape	SBD	yes	shift-invariant, sharp centres	one shape per cluster
HDBSCAN	Euclidean	no	finds rare groups, marks noise	needs aligned series
DTW k-means	DTW	yes	handles warping	slow on long or many series

SECTION 06 · COMPLETE

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